

H6 Solving differential equations A level only

Name: _____

Class: _____

Date: _____

Time: **55 minutes**

Marks: **46 marks**

Comments:

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Q1.

A student is modelling the motion of a small boat as it moves on a lake. When the speed of the boat is 12 m s^{-1} , the engine is switched off. At time t seconds later, it has a velocity of $v \text{ m s}^{-1}$ and experiences a resistance force of magnitude $20v$ newtons. The mass of the boat is 80 kg .

To set up a simple model for the motion of the boat, the student assumes that the water in the lake is still and that the boat travels in a straight line.

- (a) Explain how these two assumptions allow the student to create a simple model. (2)
- (b) State one other assumption that the student should make. (1)
- (c) (i) Express $\frac{dv}{dt}$ in terms of v . (2)
- (ii) Find an expression for v in terms of t . (5)

(Total 10 marks)

Q2.

A particle of mass 20 kg moves along a straight horizontal line. At time t seconds the velocity of the particle is $v \text{ m s}^{-1}$. A resistance force of magnitude $10\sqrt{v}$ newtons acts on the particle while it is moving. At time $t = 0$ the velocity of the particle is 25 m s^{-1} .

- (a) Show that, at time t

$$v = \left(\frac{20 - t}{4} \right)^2$$

(7)

- (b) State the value of t when the particle comes to rest.

(1)

(Total 8 marks)

Q3.

A particle travels along a straight line. At time t , the velocity of the particle is v . When $t = 0$,

$v = u$. The acceleration of the particle throughout the motion is $\frac{k}{v}$, where k is a constant.

Show that

$$v^2 = u^2 + 2kt$$

(Total 6 marks)

Q4.

John drops a rock of mass m kg from the top of a vertical cliff. At time t seconds after the rock is dropped, it has speed v m s⁻¹ and the air resistance to its motion is of magnitude $0.1mv$ newtons.

- (a) Show that the differential equation connecting v and t for this motion is $\frac{dv}{dt} = g - 0.1v$. (2)
- (b) Solve this differential equation to find v in terms of g and t . (5)
- (c) Hence show that the speed of the rock cannot exceed 98 m s⁻¹. (1)

(Total 8 marks)

Q5.

A car accelerates from rest along a straight horizontal road. It experiences a constant horizontal forward force of magnitude 2000 newtons and a resistance force. The resistance force has magnitude $40v$ newtons, when the speed of the car is v m s⁻¹. The mass of the car is 1000 kg.

- (a) Show that $\frac{dv}{dt} = \frac{50 - v}{25}$. (2)
- (b) Find the velocity of the car at time t . (5)

(Total 7 marks)

Q6.

A particle of mass m is moving along a straight horizontal line. At time t the particle has speed v . Initially the particle is at the origin and has speed U . As it moves the particle is subject to a resistance force of magnitude mkv^3 .

- (a) Show that $v^2 = \frac{U^2}{2kU^2t + 1}$. (6)
- (b) What happens to v as t increases?

(1)
(Total 7 marks)



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